

Thermodynamics of Scale: Entropic c -Functions, Data Processing, Spectral Thermodynamics, Fluctuation Relations, and the Arrow of Time

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Abstract

The irreversibility of the renormalization group (RG) is often described with thermodynamic words: information is “lost” under coarse-graining, c “decreases like an entropy”, scale behaves like a temperature, and the RG arrow is compared with the arrow of time. We sort these statements into theorems, exact identities, standard physics in a new setting, and conjecture, and we check each exact statement numerically.

First, the RG theorems (c, F, a) state that the relevant quantity *decreases* from the ultraviolet to the infrared. In the entropic proofs this follows from strong subadditivity together with Lorentz invariance, not from a thermodynamic law. For a massive lattice Dirac fermion the Casini–Huerta function $c(\ell) = 3\ell dS/d\ell$ decreases monotonically from its lattice value to zero, and at the critical point it reproduces $c = 1$ up to the finite-size chord factor $(\pi\ell/N)\cot(\pi\ell/N)$. Second, the monotonicity of relative entropy under quantum channels yields a rigorous H -theorem for any RG map that is a channel with an invariant state. It is checked on block restrictions of Gaussian fermion states. Third, if the heat-kernel time τ is read as an inverse temperature on the Laplacian spectrum, then $S(\tau) = \ln P + \tau\langle\lambda\rangle = -D(\pi_\tau\|\bar{\rho})$ is non-increasing, and the heat capacity $C = \tau^2\text{Var}(\lambda)$ obeys $C = \frac{1}{2}d_s - \frac{1}{2}dd_s/d\ln\tau$. On scaling plateaus the spectral dimension is therefore twice a heat capacity. For the hypercubic lattice C is given in closed form; it has a Schottky-type maximum $0.6799d$ and tends to the equipartition value $d/2$. Fourth, a physical protocol that changes the correlation length of a lattice field in time obeys the Jarzynski and Crooks relations exactly. We prove that the sudden-quench Jarzynski estimator has finite variance if and only if every mode stiffens by more than a factor $1/2$ in ω^2 , which explains why softening protocols fail in practice. Simulations of a stiffening protocol, combined with the Bennett acceptance ratio, recover ΔF to within 0.015.

Landauer’s bound applies to physical erasure, not to the theorist’s coarse-graining map. The identification of the thermodynamic arrow of time with the RG arrow, suggested by dS/CFT and by holographic c -theorems, is formulated as a conjecture together with what it does not explain: the low-entropy initial state.

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1 Introduction

Wilson’s renormalization group integrates out short-distance degrees of freedom, and the resulting flow in the space of theories is irreversible in a precise sense: in two dimensions Zamolodchikov’s c -theorem [1] provides a function that decreases along the flow and equals the Virasoro central charge at fixed points. In four dimensions the corresponding statement for the Euler anomaly a was conjectured by Cardy [2] and proved by Komargodski and Schwimmer [3]. In three dimensions the F -theorem was proposed in [4] and proved by Casini and Huerta from entanglement entropy [5]. In every case

$$c_{UV} > c_{IR}, \quad (1)$$

with the inequality pointing from short to long distances.

This structure invites thermodynamic language: coarse-graining “forgets” information, c counts degrees of freedom like an entropy, the RG scale is a kind of temperature, and the direction of the flow resembles the direction of time. Earlier work made parts of this precise. Gaiotto and O’Connor proposed an H -theorem for field-theory entropy [8], and Bény and Osborne described RG maps as quantum channels and measured information loss by distinguishability [9]. In holography the radial coordinate plays the role of scale [10, 11], and in dS/CFT cosmological time is dual to RG scale [12, 13].

Thermodynamic language is also easily over-extended. In this paper we separate four kinds of statements and attach a numerical check to each exact one.

1. *Theorems about the RG*, proved from quantum information inequalities (Section 2): the entropic c -function, and an H -theorem from the data-processing inequality.
2. *An exact identity* that makes “scale as inverse temperature” precise (Section 3).
3. *Standard thermodynamics applied to scale*: Landauer’s principle for physical coarse-graining devices (Section 4) and the Jarzynski and Crooks relations for protocols that change a correlation length in time (Section 5). Neither is a new law.
4. *A conjecture* that the thermodynamic arrow of time and the RG arrow coincide (Section 6).

The paper continues a series on scale-dependent structures: holographic entanglement of fractal regions [14], multifractal large deviations in entanglement [15], uncomputable fractal dimensions [16], and the crossover from discrete to continuous quantum geometry [17]. From the last of these we use the identity $d_s(\tau) = 2\tau\langle\lambda\rangle_\tau$ for the spectral dimension.

All numbers below are produced by a single script, `scale_thermodynamics.py`, whose output is reproduced in the accompanying repository. Figure 1 summarises the numerical results.

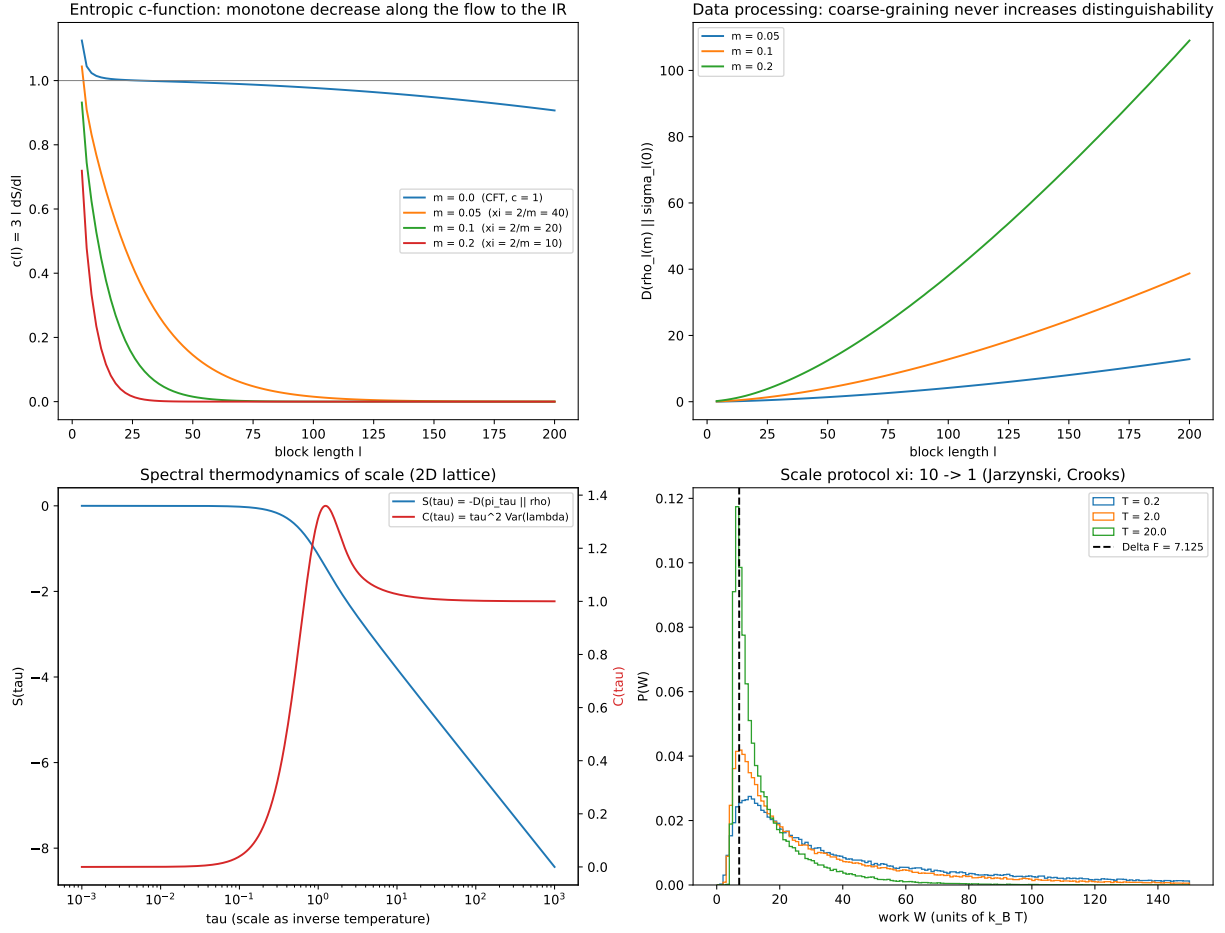


Figure 1: Four exact statements, checked numerically. *Top left:* entropic c -function $c(\ell) = 3\ell \frac{dS}{d\ell}$ of the half-filled lattice Dirac fermion ($N = 1200$) for several masses; the correlation length is $\xi = 2/m$ (Section 2.1). *Top right:* relative entropy between the massive and massless reduced states, non-decreasing in the block length as required by data processing (Section 2.2). *Bottom left:* spectral entropy $S(\tau) = -D(\pi_\tau || \bar{\rho})$ and heat capacity $C(\tau)$ of the two-dimensional square lattice (Section 3). *Bottom right:* work distributions for the protocol $m^2 : 0.01 \rightarrow 1$ (correlation length $10 \rightarrow 1$) of a Gaussian lattice field, for three durations T , with the exact ΔF (Section 5).

2 Irreversibility of the RG from entropy inequalities

2.1 The entropic c -function

Let $S(r)$ be the entanglement entropy of an interval of length r in the vacuum of a Lorentz-invariant $(1+1)$ -dimensional QFT. Casini and Huerta [6] defined

$$c(r) = 3r \frac{dS}{dr} \quad (2)$$

(their normalisation differs by the factor 3) and proved that

$$\frac{dc}{dr} = 3(rS''(r) + S'(r)) \leq 0. \quad (3)$$

At a fixed point, $S = \frac{c}{3} \ln(r/\epsilon)$ [18, 19] and $c(r)$ equals the central charge. The proof uses only two ingredients. The first is strong subadditivity (SSA) [20],

$$S(AB) + S(BC) \geq S(B) + S(ABC), \quad (4)$$

applied to intervals that are boosted relative to each other. The second is Lorentz invariance, which makes S depend only on the proper length of an interval. The irreversibility of the RG is therefore a statement about quantum correlations in the vacuum and needs no thermodynamic hypothesis. The same strategy, with the Markov property of the conformal vacuum, gives the F -theorem [5] and the a -theorem [7].

Remark. The direction of (1) is essential, and the thermodynamic analogy can mislead here. Under coarse-graining c decreases, whereas thermodynamic entropy increases. The gradient-flow form $\beta^i \partial_i c = -12 \beta^i \beta^j G_{ij}$, with a positive metric G_{ij} , is established in conformal perturbation theory [1]. In general only the inequality between fixed points and the monotonicity of a suitable c -function are proved.

Lattice check. Consider the periodic chain of N sites at half filling,

$$H = - \sum_j (c_j^\dagger c_{j+1} + \text{h.c.}) + m \sum_j (-1)^j n_j. \quad (5)$$

Its single-particle spectrum is $\pm \sqrt{4 \cos^2 k + m^2}$. The low-energy theory is a Dirac fermion with velocity $v = 2$ and mass gap m , and the correlation length is $\xi = 1/\text{arsinh}(m/2) \simeq 2/m$. Block entropies follow from the eigenvalues of the restricted correlation matrix $C_{ij} = \langle c_i^\dagger c_j \rangle$ [21] (Appendix A). We evaluate (2) by a central difference of step 2 on even block lengths, which removes the parity oscillations of the half-filled chain.

For $N = 1200$ and $\ell = 4, \dots, 200$ we find the following (Figure 1, top left).

- At $m = 0$, $c(200) = 0.907$. This matches the finite-size prediction $c(\pi\ell/N) \cot(\pi\ell/N) = 0.9069$ that follows from $S = \frac{1}{3} \ln[(N/\pi) \sin(\pi\ell/N)] + \text{const}$ [19] with $c = 1$. The initial value $c(4) = 1.125$ is a lattice effect.
- For $m = 0.05, 0.1, 0.2$, $c(\ell)$ decreases monotonically from 1.044, 0.931 and 0.719 to below 10^{-3} . The largest increment between successive points is at most 10^{-5} , i.e. at the level of round-off.
- The lattice form of SSA, $S(\ell+2) + S(\ell-2) - 2S(\ell) \leq 0$, holds for every ℓ . The largest values are -3.7×10^{-5} , -4.9×10^{-8} , -5.3×10^{-12} and $+6.8 \times 10^{-13}$ (round-off).

Remark. On a lattice, SSA and translation invariance give only the concavity of $S(\ell)$, i.e. that $c(\ell)/\ell$ is non-increasing. The monotonicity of $c(\ell)$ itself needs the Lorentz invariance of the continuum limit, so on the lattice it is a property of this model and not a theorem.

2.2 Data processing and an H -theorem for RG channels

The rigorous version of “coarse-graining loses information” is the monotonicity of the relative entropy $D(\rho\|\sigma) = \text{Tr} \rho(\ln \rho - \ln \sigma)$ under completely positive trace-preserving maps (channels).

Proposition 2.1 (Data processing [22, 23]). *For every channel Φ and all states ρ, σ , $D(\Phi(\rho)\|\Phi(\sigma)) \leq D(\rho\|\sigma)$.*

Corollary 2.1 (H -theorem for RG channels). *Let Φ be a channel on a fixed algebra of observables with an invariant state, $\Phi(\sigma_*) = \sigma_*$. Then for every initial state ρ the sequence $n \mapsto D(\Phi^n(\rho)\|\sigma_*)$ is non-increasing.*

Proof. Apply Proposition 2.1 with $\sigma = \sigma_*$: $D(\Phi^{n+1}(\rho)\|\sigma_*) = D(\Phi(\Phi^n \rho)\|\Phi(\sigma_*)) \leq D(\Phi^n \rho\|\sigma_*)$. $\square$

This is the exact sense in which an RG transformation is irreversible: if one step of the RG is a channel and the fixed point is a state invariant under it, distinguishability from the fixed point can only decrease. Scale-invariant MERA tensor networks provide examples. Their transfer (descending) maps are channels whose fixed points are the scale-invariant reduced density matrices, and the rate of convergence encodes critical exponents [24]. The corollary does not give strict decrease, and it gives a Lyapunov function relative to a chosen reference state, not an intrinsic quantity of the theory such as c . The two statements are complementary: the c -theorem needs Lorentz invariance but no reference state, and the H -theorem needs a channel and a fixed point but no symmetry.

Check. Let $\rho_\ell(m)$ be the ground state of (5) restricted to a block of ℓ sites and σ_ℓ the restriction of the massless ground state. Restriction from $\ell+2$ to ℓ sites is a partial trace, so Proposition 2.1 requires $D(\rho_\ell\|\sigma_\ell)$ to be non-decreasing in ℓ . For Gaussian states the relative entropy is given by the correlation matrices (Appendix A). We find $D(\rho_\ell\|\sigma_\ell)$ rising from 0.025, 0.072, 0.197 at $\ell = 4$ to 12.85, 38.73, 109.0 at $\ell = 200$ for $m = 0.05, 0.1, 0.2$. The smallest increments are +0.023, +0.064 and +0.17 (Figure 1, top right).

3 Spectral thermodynamics of scale

“Scale temperature” becomes exact once the heat-kernel time is read as an inverse temperature. Let Δ be a Laplacian on a graph with V vertices and eigenvalues $\lambda_1, \dots, \lambda_V$, let $\bar{\rho}$ be the uniform distribution on the spectrum, and let

$$P(\tau) = \frac{1}{V} \text{Tr} e^{-\tau \Delta}, \quad \pi_\tau(i) = \frac{e^{-\tau \lambda_i}}{V P(\tau)}, \quad \langle f \rangle_\tau = \sum_i \pi_\tau(i) f(\lambda_i). \quad (6)$$

P is the return probability of the diffusion and, at the same time, the partition function of a canonical ensemble on the spectrum at inverse temperature τ . The spectral dimension of [17] is $d_s = -2 \, d \ln P / d \ln \tau = 2\tau \langle \lambda \rangle_\tau$.

Proposition 3.1. Define $S(\tau) = \ln P(\tau) + \tau \langle \lambda \rangle_\tau$ and $C(\tau) = \tau^2 \text{Var}_\tau(\lambda)$. Then

$$S(\tau) = -D(\pi_\tau\|\bar{\rho}) = H(\pi_\tau) - \ln V \leq 0, \quad (7)$$

$$\frac{dS}{d\tau} = -\tau \text{Var}_\tau(\lambda) \leq 0, \quad (8)$$

$$C(\tau) = \frac{d_s}{2} - \frac{1}{2} \frac{dd_s}{d \ln \tau} \geq 0. \quad (9)$$

Moreover $-\ln P(\tau) = \min_p [\tau \mathbb{E}_p \lambda - H(p) + \ln V]$, with the minimum attained at $p = \pi_\tau$ (Gibbs variational principle), so $-\ln P$ is τ times a free energy.

Proof. From (6), $\ln(V \pi_\tau(i)) = -\tau \lambda_i - \ln P$, so $D(\pi_\tau\|\bar{\rho}) = \sum_i \pi_\tau(i) \ln(V \pi_\tau(i)) = -\tau \langle \lambda \rangle - \ln P = -S$; this also equals $\ln V - H(\pi_\tau)$. Next, $d \ln P / d\tau = -\langle \lambda \rangle$ and $d \langle \lambda \rangle / d\tau = -\text{Var}(\lambda)$, so $dS/d\tau = -\langle \lambda \rangle + \langle \lambda \rangle - \tau \text{Var}(\lambda)$. Finally, $dd_s/d \ln \tau = \tau d(2\tau \langle \lambda \rangle)/d\tau = d_s - 2\tau^2 \text{Var}(\lambda) = d_s - 2C$. The variational statement is the non-negativity of $D(p\|\pi_\tau) = \tau \mathbb{E}_p \lambda - H(p) + \ln V + \ln P$. $\square$

Corollary 3.1 (Dimension as heat capacity). On a scaling plateau where d_s is constant, $d_s = 2C$. In flat $\mathbb{R}^d$, $C = d/2$ exactly: each dimension contributes $1/2$, as in equipartition.

Hypercubic lattice. For $\mathbb{Z}^d$ with unit spacing, $P = (e^{-x} I_0(x))^d$ with $x = 2\tau$ [17]. Writing $r = I_1(x)/I_0(x)$ and using $r' = 1 - r/x - r^2$,

$$\langle \lambda \rangle_\tau = 2d(1 - r), \quad C(\tau) = d x^2 \left(1 - \frac{r}{x} - r^2 \right), \quad (10)$$

with $C \simeq 2d\tau^2$ as $\tau \rightarrow 0$ and $C \rightarrow d/2$ as $\tau \rightarrow \infty$. Because the factors are independent, C/d does not depend on d . Its maximum is $C = 0.6799 d$ at $\tau = 1.252$ (Figure 1, bottom left). This is a Schottky-type anomaly caused by the band edge and the bounded spectrum. By (9) it is the thermodynamic counterpart of the overshoot $\max d_s = 1.2178 d$ found in [17]. For $d = 2$ the script confirms (9) to 3×10^{-4} (finite-difference error). It also confirms $dS/d\tau \leq 0$ and the large- τ asymptote $S \simeq -\frac{d}{2} \ln(4\pi\tau) + \frac{d}{2}$, which gives $S(10^3) = -8.439$.

Remark. Proposition 3.1 is an exact dictionary, but its monotonicity is the elementary fact that a canonical ensemble loses entropy when it is cooled. It does not provide a dynamical arrow: τ is a resolution parameter, the Euclidean proper time of the diffusion, and not physical time. Its content is that the thermodynamic response of the spectrum, C , measures dimension.

4 What Landauer’s principle does and does not say

Landauer’s principle [25, 26] bounds the heat that a *physical* process must dissipate when it erases information stored in a physical memory. In the precise form of Reeb and Wolf [27], a system in state ρ_S that interacts unitarily with a bath initially in a Gibbs state at temperature T dissipates heat

$$\langle Q \rangle \geq k_B T [S(\rho_S) - S(\rho'_S)], \quad (11)$$

where ρ'_S is the final state of the system. Erasing one bit therefore costs at least $k_B T \ln 2$, which has been confirmed experimentally [28].

Three consequences follow for RG.

1. The RG of a theorist is a map between descriptions and does not act on a physical system in contact with a bath, so it costs no energy. Statements such as “every RG step dissipates $k_B T \ln 2$ per discarded degree of freedom” have no basis.
2. If a device physically implements a coarse-graining, for example a detector that records block averages and resets its fine-grained registers, then (11) applies to the device’s memory with the temperature of its bath.
3. Coarse-graining by partial trace may increase or decrease the von Neumann entropy. For example, a block of an entangled pure state has positive entropy. The relevant invariant statement is the data-processing inequality of Section 2.2, not a sign of ΔS .

5 Fluctuation relations for a physical scale protocol

The Jarzynski equality [29] and the Crooks relation [30] require a physical system driven by a time-dependent control parameter. The RG flow itself is not such a protocol. A physical protocol that changes the *scale* of a system is, however, easy to construct: change the mass, and hence the correlation length, of a field in time. The standard relations then hold, and in this sense they apply to scale.

Model. Let $\phi \in \mathbb{R}^L$ be a field on a ring with

$$H_t(\phi) = \frac{1}{2} \phi \cdot (-\Delta + m^2(t)) \phi, \quad d\phi = -\nabla H_t dt + \sqrt{2} dB_t \quad (12)$$

(overdamped Langevin dynamics, $k_B T = 1$), with $m^2(t)$ ramped linearly from m_i^2 to m_f^2 over a time T . The work is $W = \int_0^T \partial_t H_t dt = \frac{1}{2} \int_0^T \dot{m}^2 |\phi|^2 dt$. In normal modes with $q_k^2 = 4 \sin^2(\pi k/L)$ and $\omega_k^2 = q_k^2 + m^2$, the free-energy difference is

$$\Delta F = \frac{1}{2} \sum_{k=0}^{L-1} \ln \frac{q_k^2 + m_f^2}{q_k^2 + m_i^2}, \quad (13)$$

and

$$\langle e^{-W} \rangle = e^{-\Delta F}, \quad \frac{P_F(W)}{P_R(-W)} = e^{W-\Delta F}. \quad (14)$$

The lattice correlation length $\xi = 1/\text{arcosh}(1+m^2/2)$ changes from 10.0 to 1.04 for $m^2 : 0.01 \rightarrow 1$.

Variance of the estimator. The exponential average converges poorly when it is dominated by rare trajectories [31]. For quadratic Hamiltonians this can be decided exactly in the sudden limit.

Proposition 5.1. *For a sudden quench $m_i^2 \rightarrow m_f^2$ from equilibrium, the estimator e^{-W} has finite variance if and only if $R_k = \omega_{f,k}^2/\omega_{i,k}^2 > 1/2$ for every mode k . In that case its relative variance is $\prod_k R_k(2R_k - 1)^{-1/2} - 1$.*

Proof. For a sudden quench, $W = \frac{1}{2}(m_f^2 - m_i^2) \sum_k \phi_k^2$ with independent $\phi_k \sim \mathcal{N}(0, \omega_{i,k}^{-2})$. For $Z \sim \mathcal{N}(0, s^2)$, $\mathbb{E} e^{-aZ^2} = (1 + 2as^2)^{-1/2}$ if $1 + 2as^2 > 0$ and $+\infty$ otherwise. With $a = m_f^2 - m_i^2$ and $s^2 = \omega_{i,k}^{-2}$, this gives $\mathbb{E} e^{-2W} = \prod_k (2R_k - 1)^{-1/2}$ when all $R_k > 1/2$, and $+\infty$ otherwise. Dividing by $(\mathbb{E} e^{-W})^2 = \prod_k R_k^{-1}$ gives the relative variance. $\square$

For our ring ($L = 16$) the forward (stiffening) protocol has $\min_k R_k = 1.25$ and a sudden-limit relative variance of 73.6. The reverse (softening) protocol has $R_0 = 0.01 < 1/2$, so its estimator has infinite variance. This is why a first attempt that used the softening direction as the forward protocol gave estimates biased by more than one unit. Estimates of ΔF should therefore be taken from the stiffening direction, or from the Bennett acceptance ratio [32, 33], which combines both directions optimally.

Simulation. We integrate (12) exactly, mode by mode, using the Ornstein–Uhlenbeck propagator over 400 steps, for $M = 10^5$ trajectories in each direction. The exact value is $\Delta F = 7.1254$. Table 1 shows that the Jarzynski estimates agree with it within one standard error, that the Crooks slope is 1 to within 2%, and that the Bennett estimate agrees to within 0.008. The dissipated work $\langle W \rangle - \Delta F$ is positive and decreases towards the quasi-static limit, as the second law requires (Figure 1, bottom right).

Table 1: Scale protocol $m^2 : 0.01 \rightarrow 1$ on a ring of $L = 16$ sites, $M = 10^5$ trajectories per direction. Exact $\Delta F = 7.1254$. The Crooks slope is the fitted slope of $\ln[P_F(W)/P_R(-W)]$ against W , expected to be 1.

T	$\langle W \rangle$	$\langle W \rangle - \Delta F$	Jarzynski	Crooks slope	Bennett
0.2	55.68	48.56	7.140 ± 0.023	0.995	7.131
2	36.76	29.63	7.139 ± 0.015	0.981	7.133
20	15.81	8.69	7.126 ± 0.007	1.003	7.122

6 The RG arrow and the arrow of time

The statements so far are theorems or exact identities. The relation between the RG arrow and the arrow of time is of a different kind, and we state what is established before formulating a conjecture.

Established. (i) The RG is irreversible in the sense of Sections 2.1 and 2.2. (ii) In holographic RG flows the radial coordinate of the bulk acts as the scale. A c -function built from the warp factor is monotone whenever the bulk matter satisfies the null energy condition [10, 11]. (iii) In dS/CFT, if it exists, the late-time boundary of de Sitter space carries a Euclidean CFT, and cosmological time evolution corresponds to inverse RG flow [12, 13]. Expansion of the universe then corresponds to flow towards the UV of the dual theory.

Conjecture 1 (RG arrow = thermodynamic arrow). *In a cosmology with a holographic dual of the dS/CFT type, the direction in which the coarse-grained entropy of the bulk increases coincides with a monotone direction of an RG c -function of the dual theory.*

What the conjecture would not explain. Even if Conjecture 1 were proved, the thermodynamic arrow of time would rest on the low-entropy initial state of the universe, the *Past Hypothesis* [34]. The conjecture would move that boundary condition to the dual theory, as a condition on which endpoint of the flow corresponds to the past, but it would not derive it. Moreover, microscopic dynamics is time-reversal invariant (up to CPT), whereas the c -theorem concerns a flow in the space of theories, not the evolution of states. A precise version of the conjecture must therefore specify which bulk entropy is meant and how the boundary condition at one end of the flow is chosen. dS/CFT itself is not established: the putative dual theories are non-unitary. For these reasons we regard Conjecture 1 as a research direction, not as a result.

7 Conclusion

The thermodynamics of scale has a rigorous core that is information-theoretic rather than thermodynamic.

- RG irreversibility, $c_{UV} > c_{IR}$, follows from strong subadditivity and Lorentz invariance.
- Data processing gives an H -theorem for any RG channel with an invariant state.
- Heat-kernel time is an exact inverse temperature for the spectrum. The heat capacity C then satisfies $C = \frac{1}{2}d_s - \frac{1}{2}dd_s/d\ln\tau$, so dimension is twice a heat capacity on scaling plateaus.

Thermodynamics proper enters only when scale is changed physically. Landauer’s principle bounds devices that erase fine-grained records, and the Jarzynski and Crooks relations hold for protocols that change a correlation length in time. For these protocols we gave an exact criterion for when the Jarzynski estimator converges. The identification of the RG arrow with the arrow of time remains a conjecture, and it does not explain the low-entropy past.

Data and code availability

The script `scale_thermodynamics.py`, the data (CSV), the figures and the console output are available at <https://github.com/Ruqing1963/scale-thermodynamics> and archived under DOI 10.5281/zenodo.23199122.

A Numerical methods

Gaussian fermion states. For a quadratic Hamiltonian the reduced state of a block A is Gaussian and fixed by $C_A = (\langle c_i^\dagger c_j \rangle)_{i,j \in A}$ [21]. With eigenvalues ν_k of C_A , $S = -\sum_k [\nu_k \ln \nu_k +$

$(1 - \nu_k) \ln(1 - \nu_k)]$. If $\sigma \propto \exp(-\sum_{ij} h_{ij} c_i^\dagger c_j)$ has correlation matrix G , then $h = \ln[(1 - G)G^{-1}]$ (matrix logarithm), $\ln Z_\sigma = -\text{Tr} \ln(1 - G)$, and

$$D(\rho \parallel \sigma) = -S(\rho) + \text{Tr}(hC) - \text{Tr} \ln(1 - G) \quad (15)$$

for real symmetric correlation matrices. Eigenvalues are clipped to $[10^{-15}, 1 - 10^{-15}]$.

Ornstein–Uhlenbeck integration. Between switching steps the parameter is frozen and each mode evolves exactly, $\phi_k \mapsto a\phi_k + \sqrt{(1 - a^2)/\omega_k^2} \xi$ with $a = e^{-\omega_k^2 \Delta t}$ and $\xi \sim \mathcal{N}(0, 1)$. The work increment at a switch is $\frac{1}{2} \Delta m^2 |\phi|^2$. This discretisation satisfies the Jarzynski equality exactly for every step size, so the only errors are statistical.

Estimators. By the delta method, the standard error of the Jarzynski estimate is $s/(\sqrt{M} \mu)$, where μ and s are the sample mean and standard deviation of e^{-W} . The Crooks slope is fitted over the bins in which both histograms contain more than 50 counts. For equal sample sizes the Bennett estimate solves

$$\sum_F [1 + e^{W_F - \Delta F}]^{-1} = \sum_R [1 + e^{W_R + \Delta F}]^{-1}.$$

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